Flow Matching for Generative Modeling

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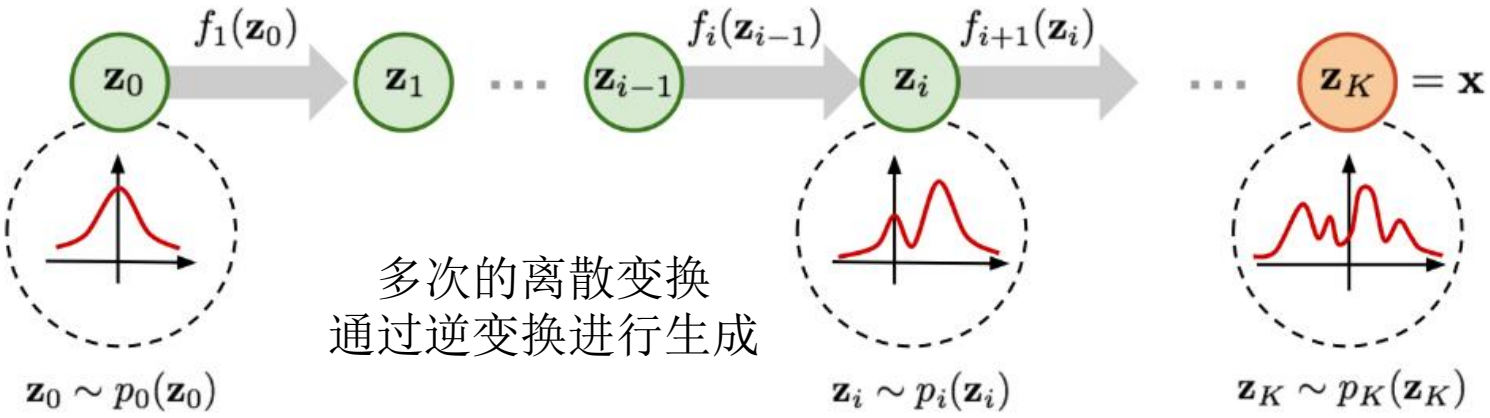
Preliminaries

- Normalizing Flows



多次的离散变换
通过逆变换进行生成

1. 已知随机变量z和其概率密度函数$z \sim \pi(z)$
2. 可逆映射$x = f(z)$以及$z = f^{-1}(x)$
3. $\int p(x)dx = \int \pi(z)dz = 1$
4. $\boldsymbol{p(x)} = \pi(z)\left|\frac{dz}{dx}\right| = \pi\left(f^{-1}(x)\right)\left|\frac{df^{-1}}{dx}\right| = \boldsymbol{\pi\left(f^{-1}(x)\right)|det\, J(f^{-1}(x))|}$ $\quad (*)$ $\qquad p(x) = [f]_* \pi(z)$

- Continuous Normalizing Flow 速度场，定义数据点在t时刻的变化大小和方向

$$\frac{dz_t}{dt} = v_t(z_t)$$

连续性方程(v_t的必要约束)

$$\frac{d\pi_t(z_t)}{dt} = -\nabla \cdot (\pi_t(z_t) v_t(z_t))$$

可通过数值计算求解分布 $\longrightarrow$

$$z_{t+\Delta t} = z_t + v_t(z_t)\Delta t$$

论文中的问题定义：$x = [x^1, \ldots, x^d]$, $\quad t \in [0,1]$ $\quad p_t$: t时刻分布

$$\frac{d\phi_t(x)}{dt} = v_t(\phi_t(x)) \qquad \phi_0(x) = x$$

ϕ_t变换算子(流)

$$p_t(x) = [\phi_t]_* p_0(x)$$

$$[\phi_t]_* p_0(x) = p_0\left(\phi_t^{-1}(x)\right) \det\left[\frac{\partial \phi_t^{-1}}{\partial x}(x)\right]$$

Background

生成模型：已知分布->目标分布，**建模已知分布到目标分布的变换**

- Diffusion Model

$$\text{已知分布} \underset{\text{加噪}}{\overset{\text{去噪}}{\rightleftarrows}} \text{目标分布}$$

概率路径单一，训练时间长

- Continuous Normalizing Flow(CNF)

$$\frac{d\phi_t(x)}{dt} = v_t(\phi_t(x))$$

$$[\phi_t]_* p_0(x) = p_0\left(\phi_t^{-1}(x)\right) \det\left[\frac{\partial \phi_t^{-1}}{\partial x}(x)\right]$$

可以建模任意概率路径

没有好的训练方法

需要昂贵的ODE模拟

Target 1.设计无模拟、高效的CNF训练方法
2.打破扩散模型的路径限制

Flow Matching(FM)

$$\frac{d\phi_t(x)}{dt} = v_t(\phi_t(x))$$

拟合$v_t(x)$ ⟶ 考虑$v_\theta(t,x): [0,1] \times R_d \to R_d$ 那么：

$$\boxed{L_{FM}(\theta) = E_{t,p_t(x)}||v_\theta(t,x) - v_t(x)||^2}$$

Flow Matching的loss

$v_t(x)$未知？如何找代理目标？考虑条件分布形式

$$p_t(\mathrm{x}) = \int p_t(x|z)q(z)dz \xrightarrow{\text{连续性方程}} \frac{\partial p_t(x|z)}{\partial t} = -\nabla \cdot (p_t(x|z)v_t(x|z))$$

$$\frac{\partial p_t(x)}{\partial t} = -\nabla \cdot (p_t(x)v_t(x))$$

$$\frac{\partial p_t(x)}{\partial t} = -\frac{\partial}{\partial_t}\int p_t(x|z)q(z)dz = -\int \left[\frac{\partial}{\partial_t}p_t(x|z)\right]q(z)dz = -\nabla \cdot \int v_t(x|z)p_t(x|z)q(z)dz = -\nabla \cdot (p_t(x)v_t(x))$$

$$\Downarrow$$

$$v_t(x) = \int \frac{v_t(x|z)p_t(x|z)q(z)}{p_t(x)}dz$$

有何意义？

通过条件分布$v_t(x|z)$（对应生成$p_t(x|z)$）
可以构造边际分布$v_t(x)$，生成$p_t(x)$

Conditional Flow Matching(CFM)

Flow Matching的loss

处理积分？

$$L_{FM}(\theta) = E_{t,p_t(x)}||v_\theta(t,x) - v_t(x)||^2$$

直觉

Conditional Flow Matching的loss

$$L_{CFM}(\theta) = E_{t,p_t(x|z),q(z)}||v_\theta(t,x) - v_t(x|z)||^2$$

考查梯度

Flow Matching

$$\nabla_\theta L_{FM}(\theta) = \nabla_\theta E_{t,p_t(x)}||v_\theta(t,x) - v_t(x)||^2$$

$$= \nabla_\theta E_{t,p_t(x)}[v_\theta(t,x)]^2 - 2\nabla_\theta E_{t,p_t(x)}[v_\theta(t,x) \cdot v_t(x)]$$

$$\nabla_\theta E_{t,p_t(x)}[v_\theta(t,x) \cdot v_t(x)]$$

$$= \nabla_\theta \iint v_\theta(t,x) v_t(x) p_t(x) dt dx = \nabla_\theta \iint v_\theta(t,x) \left[\int \frac{v_t}{\cdot v_t(x|z)}\right]$$

Conditional Flow Matching

$$\nabla_\theta L_{CFM}(\theta) = \nabla_\theta E_{t,p_t(x),q(z)}||v_\theta(t,x) - v_t(x|z)||^2$$

$$= \nabla_\theta E_{t,p_t(x|z),q(z)}[v_\theta(t,x)]^2 - 2\nabla_\theta E_{t,p_t(x|z),q(z)}[v_\theta(t,x) \cdot v_t(x|z)]$$

$$\nabla_\theta L_{FM}(\theta) = \nabla_\theta L_{CFM}(\theta)$$

Conditional Flow Matching(CFM)

目标:$v_t(x|z)$

t=0时标准正态分布，t=1时，为样本z附近的正态

考虑条件分布:$p_t(x|z) = \mathcal{N}(x|\mu_t(z), \sigma_t(z)^2 I)$, $\mu_0(z) = 0,\ \sigma_0(z) = 1$, $\mu_1(z) = z,\ \sigma_1(z) = \sigma_{min}$

设计满足条件分布的概率流（有无穷多条）

考虑条件概率流:$\phi_t(x) = \sigma_t(z)x + \mu_t(z)$

$$[\phi_t]_* p_0(x|z) = p_t(x|z)$$

$$\frac{d\phi_t(x)}{dt} = v_t(\phi_t(x)|z)$$

设服从$p_t(x|z)$ 的随机变量为y

$$y = \phi_t(x),\ \ x = \phi_t^{-1}(y)$$

$$\phi_t'(\phi_t^{-1}(y)) = v_t(y|z) \quad (*)$$

$$\phi_t^{-1}(y) = \frac{y - \mu_t(z)}{\sigma_t(z)} \quad (1)$$

$$\phi_t'(x) = \sigma_t'(z)x + \mu_t'(z) \quad (2)$$

代入（$*$）

$$\boxed{v_t(y|z) = \frac{\sigma_t'(z)}{\sigma_t(z)}\big(y - \mu_t(z)\big) + \mu_t{}'(z)}$$

总结：

（1）自主构造初始分布和目标分布之间的概率流得到μ_t，σ_t

（2）计算$v_t(y|z)$

（3）可代入$L_{CFM}(\theta)$训练

Special Instances

$$v_t(x|z) = \frac{\sigma_t'(z)}{\sigma_t(z)}\big(x - \mu_t(z)\big) + \mu_t{}'(z)$$

- Variance Exploding (VE) path 方差爆炸路径

$$x_t = x_0 + \sigma_t \epsilon$$

$$p_t(x|x_0) = \mathcal{N}(x|x_0, \sigma_t^2 I) \quad \mu_t(x_0) = x_0,\ \ \sigma_t(x_0) = \sigma_t$$

$$v_t(x|x_0) = \frac{\sigma_t'}{\sigma_t}(x - x_0)$$

- Variance Preserving (VP) path 方差保持路径

$$x_t = \alpha_t x_0 + \sigma_t \epsilon \qquad \alpha_t^2 + \sigma_t^2 = 1$$

$$p_t(x|x_0) = \mathcal{N}(x|\alpha_t x_0, \sigma_t^2 I) \quad \mu_t(x_0) = \alpha_t x_0,\ \ \sigma_t(x_0) = \sigma_t$$

$$v_t(x|x_0) = \frac{\sigma_t'}{\sigma_t}(x - \alpha_t x_0) + \alpha_t' x_0$$

（这里的x_0是样本，不是时间（原文为x_1），为了和DDPM等对齐）

- Optimal Transport(OT) path 最优传输路径

$$\mu_t(x) = t x_1,\ \ \sigma_t(x) = 1 - (1 - \sigma_{min})t$$

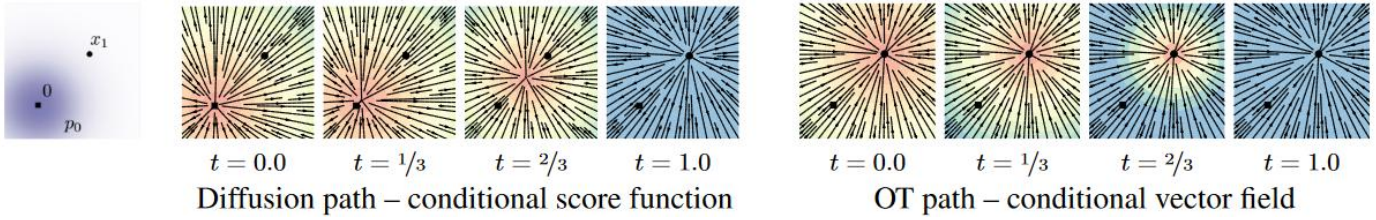
$$v_t(x|x_1) = \frac{x_1 - (1 - \sigma_{min})x}{1 - (1 - \sigma_{min})t}$$

$$x = \phi_t(x_0) = (1 - (1 - \sigma_{min})t)x_0 + t x_1$$

$$v_t(x|x_0, x_1) = x_1 - (1 - \sigma_{min})x_0$$

$$L_{CFM}(\theta) = E_{t,q(x_1),p(x_0)}||v_t(\phi_t(x_0)) - (x_1 - (1 - \sigma_{min})x_0)||^2$$

给定x_0，速度不变（直线）

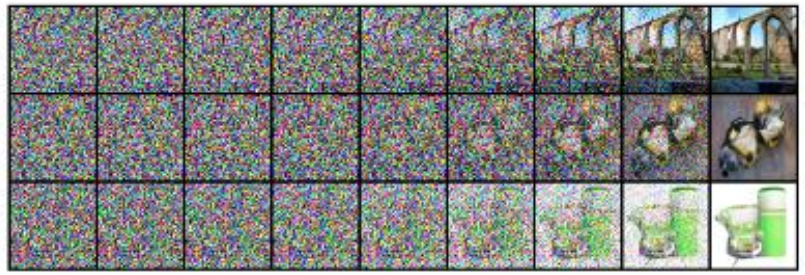


Experiments

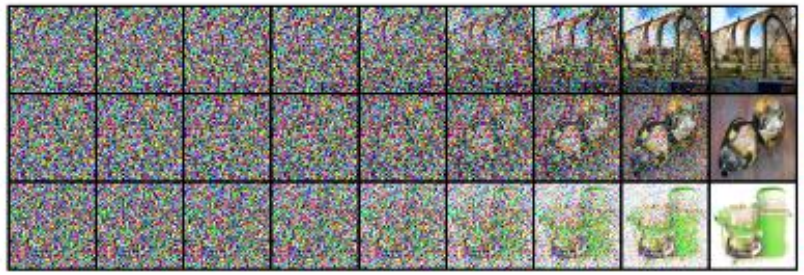
CIFAR-10、ImageNet（32×32、64×64、128×128、256×256）

Model	CIFAR-10			ImageNet 32×32			ImageNet 64×64		
	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓
Ablations									
DDPM	3.12	7.48	274	3.54	6.99	262	3.32	17.36	264
Score Matching	3.16	19.94	242	3.56	5.68	178	3.40	19.74	441
ScoreFlow	3.09	20.78	428	3.55	14.14	195	3.36	24.95	601
Ours									
FM w/ Diffusion	3.10	8.06	183	3.54	6.37	193	3.33	16.88	187
FM w/ OT	**2.99**	**6.35**	**142**	**3.53**	**5.02**	**122**	**3.31**	**14.45**	**138**

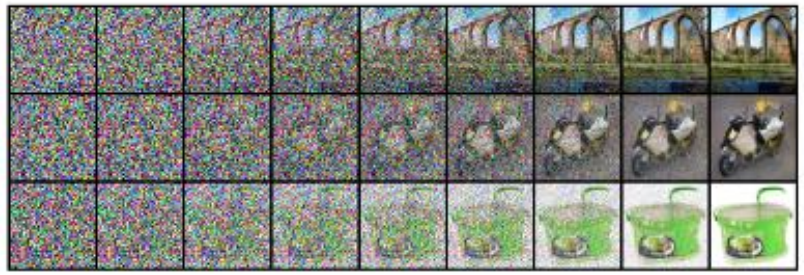
Model	ImageNet 128×128	
	NLL↓	FID↓
MGAN (Hoang et al., 2018)	–	58.9
PacGAN2 (Lin et al., 2018)	–	57.5
Logo-GAN-AE (Sage et al., 2018)	–	50.9
Self-cond. GAN (Lučić et al., 2019)	–	41.7
Uncond. BigGAN (Lučić et al., 2019)	–	25.3
PGMGAN (Armandpour et al., 2021)	–	21.7
FM w/ OT	**2.90**	**20.9**



Score Matching w/ Diffusion



Flow Matching w/ Diffusion

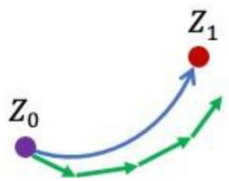


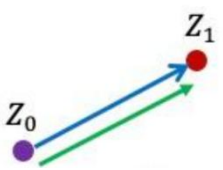
Flow Matching w/ OT

Rectified Flow

$$\frac{dz_t}{dt} = v_t(z_t) \qquad z_{t+\Delta t} = z_t + v_t(z_t)\Delta t$$

走直线可以让步长取得大，加速生成





问题setting

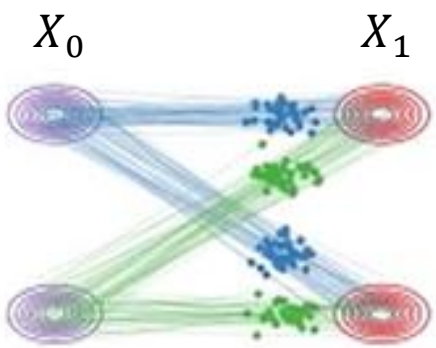
$x_0 \sim \pi_0$，$x_1 \sim \pi_1$　把x_0和x_1用线性插值连起来 $\longrightarrow$ $x_t = tx_1 + (1-t)x_0$，$t \in [0,1]$

求导得到速度　$\frac{dx_t}{dt} = v_t(x_t) = x_1 - x_0$　并非因果系统（推理时用到未来信息x_1）

尝试用模型$v_\theta(x_t, t)$拟合

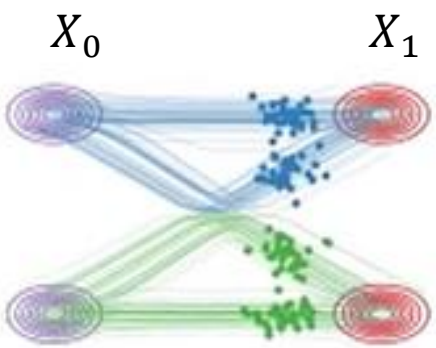
$$L = \int_0^1 E_{x_0 \sim \pi_0, x_1 \sim \pi_1} \left[\left|\left| (x_1 - x_0) - v_\theta(x_t, t) \right|\right|^2 \right] dt$$

x_0和x_1的配对



$v_t(x_t) = x_1 - x_0$

非因果系统



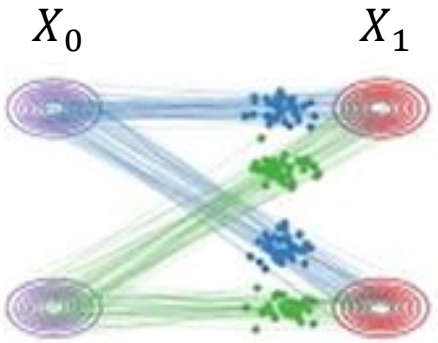
$v_t(x_t) = v_\theta(x_t, t)$

因果系统

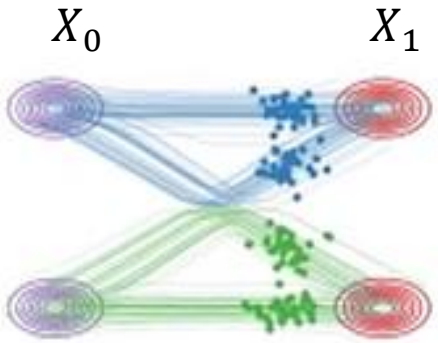
由于因果系统，学习得到的模型还是不能走直线！

Rectified Flow

$$L = \int_0^1 E_{x_0 \sim \pi_0, x_1 \sim \pi_1} \left[\left\| (x_1 - x_0) - v_\theta(x_t, t) \right\|^2 \right] dt$$



$v_t(x_t) = x_1 - x_0$



$v_t(x_t) = v_\theta(x_t, t)$

$$L = \int_0^1 E_{x_0 \sim \pi_0, x_1 \sim \pi_1} \left[\left\| (x_1 - x_0) - v_\theta(x_t, t) \right\|^2 \right] dt$$

$$= \int_0^1 E_{x_0, x_1} \left[\|(x_1 - x_0)\|^2 + \|v_\theta(x_t, t)\|^2 - 2(x_1 - x_0) v_\theta(x_t, t) \right] dt$$

$$= \int_0^1 E_{x_0, x_1} \left[\|v_\theta(x_t, t)\|^2 \right] - 2E_{x_0, x_1}[(x_1 - x_0) v_\theta(x_t, t)]\, dt + c$$

$$= \int_0^1 E_{x_t} \left[\|v_\theta(x_t, t)\|^2 \right] - 2E_{x_t}[E_{x_0, x_1}[(x_1 - x_0)|x_t] E_{x_t}[v_\theta(x_t, t)]] dt + c$$

$$= \int_0^1 E_{x_t} \left[\left\| E_{x_0, x_1}[x_1 - x_0 | x_t] - v_\theta(x_t, t) \right\|^2 \right] dt + c'$$

Rectified Flow的本质：
将非因果的ODE

$$\frac{dx_t}{dt} = v_t(x_t) = x_1 - x_0$$

学习为因果的ODE

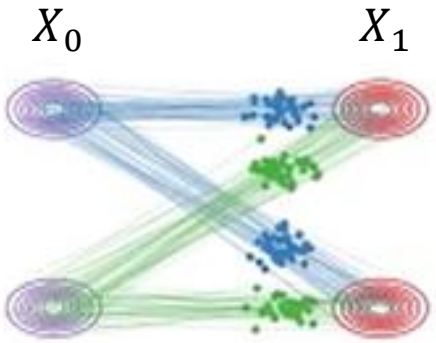
$$\frac{dx_t}{dt} = v_t(x_t) = E_{x_0, x_1}[x_1 - x_0 | x_t]$$

而且保证数据分布不变

最优目标为$\boxed{v_\theta^*(x_t, t) = E_{x_0, x_1}[x_1 - x_0 | x_t]}$ 是一个期望，不一定能走直线

Rectified Flow---Reflow!

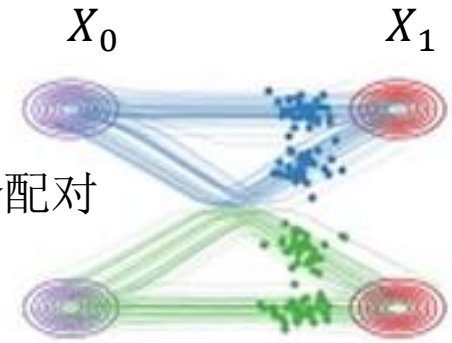
随机配对



$v_t(x_t) = x_1 - x_0$

非因果系统

1Flow配对



$v_t(x_t) = v_\theta(x_t, t)$

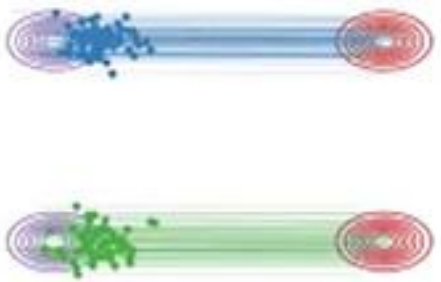
因果系统

从π_0中采样x_0, 从π_1中采样x_1

不再从π_1中采样x_1

改为从$Flow_1(x_0)$中算出x_1, 天然存在数据配对

拉直轨迹



$$L = \int_0^1 E_{x_0 \sim \pi_0, x_1 = Flow_1(x_0)} \left[\left|\left|(x_1 - x_0) - v_\theta(x_t, t)\right|\right|^2 \right] dt$$

2-Rectified Flow

（1）理论上可以做3-Rectified Flow 4-Rectified Flow等

（2）等价于减小最优传输理论中的传输代价

（3）但是会累计误差(用精度换速度)

Reflow VS Distillation(蒸馏)

Distillation(蒸馏)能否一步到位？

$$x_1 = x_0 + v(x_0, 0)$$

$$L = E_{x_0 \sim \pi_0, x_1 \sim \pi_1}\left[\left|\left|(x_1 - x_0) - v_\theta(x_0, 0)\right|\right|^2\right]$$

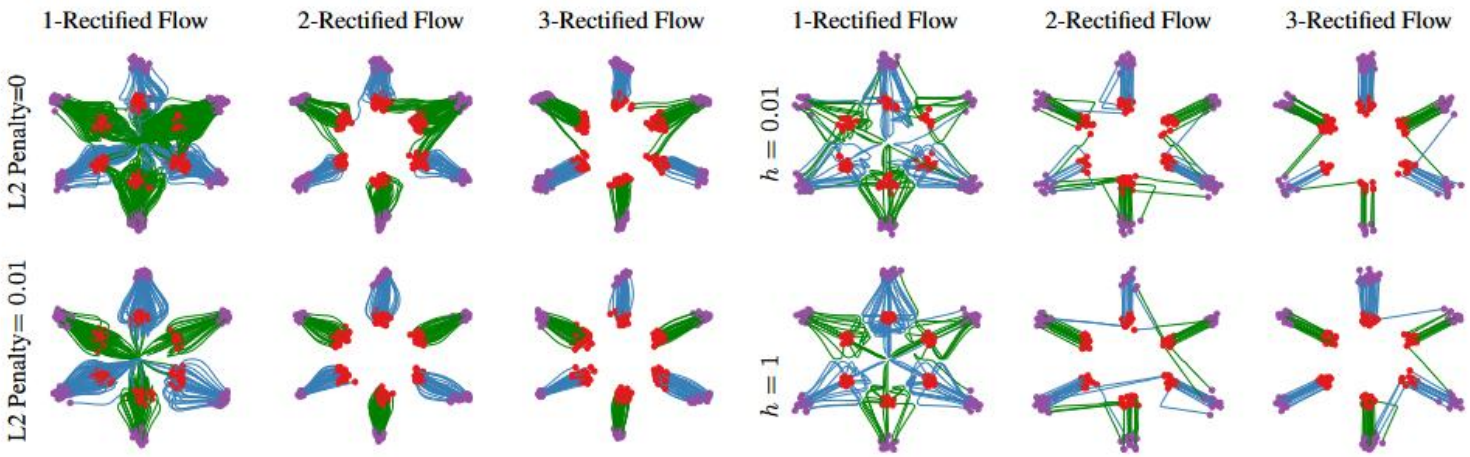
$$v_\theta^*(x_0, 0) = E_{x_0, x_1}[x_1 - x_0 | x_0]$$

一步蒸馏学出来的是条件平均

Reflow

拥有更好的数据配对(x_1, x_0)的先验

$$L = \int_0^1 E_{x_0 \sim \pi_0, x_1 = Flow_1(x_0)}\left[\left|\left|(x_1 - x_0) - v_\theta(x_t, t)\right|\right|^2\right] dt$$



Experiments

Method	NFE(↓)	IS (↑)	FID (↓)	Recall (↑)
ODE	*One-Step Generation (Euler solver, N=1)*			
1-Rectified Flow (+*Distill*)	1	1.13 (***9.08***)	378 (*6.18*)	0.0 (*0.45*)
2-Rectified Flow (+*Distill*)	1	8.08 (*9.01*)	12.21 (***4.85***)	0.34 (*0.50*)
3-Rectified Flow (+*Distill*)	1	8.47 (*8.79*)	8.15 (*5.21*)	0.41 (***0.51***)
VP ODE [73] (+*Distill*)	1	1.20 (*8.73*)	451 (*16.23*)	0.0 (*0.29*)
sub-VP ODE [73] (+*Distill*)	1	1.21 (*8.80*)	451 (*14.32*)	0.0 (*0.35*)
ODE	*Full Simulation (Runge–Kutta (RK45), Adaptive N)*			
1-Rectified Flow	127	**9.60**	**2.58**	**0.57**
2-Rectified Flow	110	9.24	3.36	0.54
3-Rectified Flow	104	9.01	3.96	0.53
VP ODE [73]	140	9.37	3.93	0.51
sub-VP ODE [73]	146	9.46	3.16	0.55
SDE	*Full Simulation (Euler solver, N=2000)*			
VP SDE [73]	2000	9.58	2.55	0.58
sub-VP SDE [73]	2000	9.56	2.61	0.58

(a) Results using the DDPM++ architecture.

Method	NFE(↓)	IS (↑)	FID (↓)	Recall (↑)
GAN	*One-Step Generation*			
SNGAN [52]	1	8.22	21.7	0.44
StyleGAN2 [28]	1	9.18	8.32	0.41
StyleGAN-XL [66]	1	-	1.85	0.47
StyleGAN2 + ADA [28]	1	9.40	2.92	0.49
StyleGAN2 + DiffAug [98]	1	9.40	5.79	0.42
TransGAN + DiffAug [26]	1	9.02	9.26	0.41
GAN with U-Net	*One-step Generation*			
TDPM (T=1) [99]	1	8.65	8.91	0.46
Denoising Diffusion GAN (T=1) [91]	1	8.93	14.6	0.19
ODE	*One Step Generation (Euler solver, N=1)*			
DDIM Distillation [47]	1	8.36	9.36	0.51
NCSN++ (VE ODE) [73] (+*Distill*)	1	1.18 (*2.57*)	461 (*254*)	0.0 (*0.0*)
ODE	*Full Simulation (Runge–Kutta (RK45), Adaptive N)*			
NCSN++ (VE ODE) [73]	176	9.35	5.38	0.56
SDE	*Full Simulation (Euler solver)*			
DDPM [23]	1000	9.46	3.21	0.57
NCSN++ (VE SDE) [73]	2000	9.83	2.38	0.59

(b) Recent results with different architectures reported in literature.

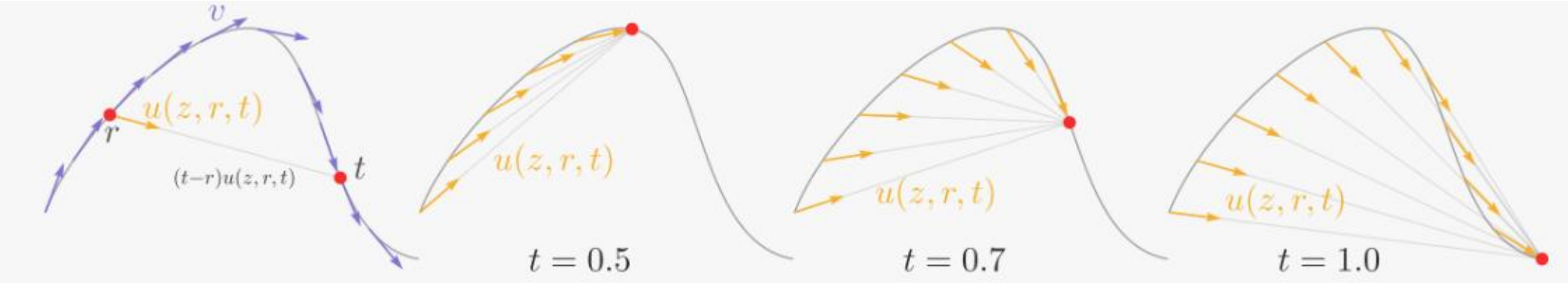
可以显著降低推理步数，可以采取Reflow+蒸馏的策略

Mean Flow

$\frac{dz_t}{dt} = v_t(z_t)$ $\quad z_{t+\Delta t} = z_t + v_t(z_t)\Delta t$ 还是依赖瞬时速度，数值计算 $\longrightarrow$ 瞬时速度改为平均速度？

$$u(z_t, r, t) = \frac{1}{t-r}\int_r^t v(z_\tau, \tau)d\tau$$

$z_1 = z_0 + u_\theta(z_0, 0,1) \longrightarrow$ 一步生成



Mean Flow

$$u(z_t, r, t) = \frac{1}{t-r}\int_r^t v(z_\tau, \tau)d\tau$$

$$(t-r)u(z_t, r, t) = \int_r^t v(z_\tau, \tau)d\tau$$ 两边关于t求导数

$$u + (t-r)\frac{du}{dt} = v(z_t, t) \longrightarrow \boxed{u = v(z_t, t) - (t-r)\frac{du}{dt}}$$

其中$\frac{du}{dt} = \frac{\partial u}{\partial z}\frac{dz_t}{dt} + \frac{\partial u}{\partial r}\frac{dz_r}{dt} + \frac{\partial u}{\partial t}\frac{dt}{dt} = \frac{\partial u}{\partial z}v(z_t, t) + \frac{\partial u}{\partial t}$

$$L(\theta) = E\left[\left\|u_\theta(z_t, r, t) - \left(v(z_t, t) - (t-r)\frac{du}{dt}\right)\right\|^2\right]$$

$$\downarrow$$

$$L(\theta) = E\left[\left\|u_\theta(z_t, r, t) - \left(v(z_t, t) - (t-r)\frac{du_\theta}{dt}\right)\right\|^2\right]$$

Algorithm 1 MeanFlow: Training.

Note: in PyTorch and JAX, `jvp` returns the function output and JVP.

```
# fn(z, r, t): function to predict u
# x: training batch

t, r = sample_t_r()
e = randn_like(x)

z = (1 - t) * x + t * e
v = e - x

u, dudt = jvp(fn, (z, r, t), (v, 0, 1))

u_tgt = v - (t - r) * dudt
error = u - stopgrad(u_tgt)

loss = metric(error)
```

Thanks!

Rectified Flow（附）

Rectified Flow的本质：
将非因果的ODE

$$\frac{dx_t}{dt} = v_t(x_t) = x_1 - x_0$$

学习为因果的ODE

$$\frac{dx_t}{dt} = v_t(x_t) = E_{x_0,x_1}[x_1 - x_0|x_t]$$

而且保证数据分布不变？

原始数据$X_0 \sim \pi_0, X_1 \sim \pi_1$, $X_t = tX_1 + (1-t)X_0$, $v^*(x_t, t) = E[x_1 - x_0|x_t]$

整流数据$Z_0 = X_0, Z_t = Z_0 + \int_0^t v^*(z_\tau, \tau)d\tau$

$Law(X_t) = Law(Z_t)$?

$$\frac{d}{dt}E[h(X_t)] = E[\nabla h(X_t)^T X_t'] = E[\nabla h(X_t)^T E[X_t'|X_t]] = E[\nabla h(X_t)^T v^*(X_t, t)]$$

$$\frac{d}{dt}E[h(Z_t)] = E[\nabla h(X_t)^T Z_t'] = E[\nabla h(X_t)^T v^*(Z_t, t)]$$

满足同一个连续性方程

Rectified Flow（附）

传输代价减小的证明

$$
\begin{aligned}
\mathbb{E}\left[c(Z_1 - Z_0)\right] &= \mathbb{E}\left[c\left(\int_0^1 v^{\boldsymbol{X}}(Z_t, t)\mathrm{d}t\right)\right] && \text{//as } \mathrm{d}Z_t = v^{\boldsymbol{X}}(Z_t, t)\mathrm{d}t \\
&\leq \mathbb{E}\left[\int_0^1 c\left(v^{\boldsymbol{X}}(Z_t, t)\right)\mathrm{d}t\right] && \text{//convexity of } c\text{, Jensen's inequality} \\
&= \mathbb{E}\left[\int_0^1 c\left(v^{\boldsymbol{X}}(X_t, t)\right)\mathrm{d}t\right] && \text{//} X_t \text{ and } Z_t \text{ shares the same marginals} \\
&= \mathbb{E}\left[\int_0^1 c\left(\mathbb{E}\left[(X_1 - X_0) \mid X_t\right]\right)\mathrm{d}t\right] && \text{//definition of } v^{\boldsymbol{X}} \\
&\leq \mathbb{E}\left[\int_0^1 \mathbb{E}\left[c\left(X_1 - X_0\right) \mid X_t\right]\mathrm{d}t\right] && \text{//convexity of } c\text{, Jensen's inequality} \\
&= \int_0^1 \mathbb{E}\left[c\left(X_1 - X_0\right)\right]\mathrm{d}t && \text{//}\mathbb{E}[\mathbb{E}[(X_1 - X_0)|X_t]] = \mathbb{E}[(X_1 - X_0)] \\
&= \mathbb{E}\left[c\left(X_1 - X_0\right)\right].
\end{aligned}
$$